

Regression Analysis Lab Report: California Housing

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1. Exploratory Data Analysis (EDA)

The dataset was explored using summary statistics, histograms, scatter plots, and a correlation matrix. The dataset contains 20,640 observations and 9 variables, including 8 predictors and the target variable MedHouseVal, which represents the median house value.

Correlation analysis shows that median income (MedInc) has the strongest positive relationship with house prices, with a correlation of 0.688. This indicates that areas with higher median income tend to have significantly higher house values.

Other variables show weaker relationships with the target. For example AveRooms (0.152) and HouseAge (0.106) have weak positive correlations. Latitude (-0.144) and Longitude (-0.046) show slight negative correlations, reflecting geographic influences on housing prices.

Histograms indicate that several variables such as Population and AveOccup are right-skewed, meaning most observations have relatively small values while a few observations have very large values. Scatter plots confirm that MedInc is the most important predictor of house prices.

2. Multiple Regression Model (Initial)

A multiple linear regression model was built using all available predictors including MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, and Longitude.

The regression coefficients show how the predicted house value changes when each predictor increases by one unit while holding other variables constant. The model indicates that MedInc has the strongest positive effect on house prices, confirming the findings from the correlation analysis.

3. Model Refinement

To assess multicollinearity between predictors, the Variance Inflation Factor (VIF) was calculated. Most predictors have acceptable VIF values below 10. However, several variables show moderate multicollinearity.

AveRooms (VIF ≈ 8.34) and AveBedrms (VIF ≈ 6.99) are highly related because houses with more rooms typically have more bedrooms. Latitude (VIF ≈ 9.29) and Longitude (VIF ≈ 8.96) are also correlated because they both describe geographic location.

Although some predictors show moderate correlation, all VIF values remain below the commonly used threshold of 10, so all variables were retained in the model. Residual plots show that prediction errors are randomly distributed around zero, suggesting that the assumptions of linear regression are reasonably satisfied. The histogram of residuals indicates an approximately normal distribution of errors.

4. Interpret Results

The regression results indicate that median income (MedInc) is the most important predictor of housing prices. The coefficient for MedInc is approximately 0.449, meaning that when median income increases

by one unit (about \$10,000 in this dataset), the predicted house value increases by approximately 0.449 units of MedHouseVal.

Since MedHouseVal is measured in hundreds of thousands of dollars, this corresponds to an approximate increase of about \$44,900 in the median house price.

HouseAge has a small positive influence on house values. AveRooms and AveBedrms show mixed effects due to their correlation with each other. Population and AveOccup have very small impacts on house prices, while Latitude and Longitude indicate that geographic location also influences house values.

5. Prediction and Model Evaluation

To evaluate the model's predictive performance, the dataset was split into training (80%) and testing (20%) sets. The model achieved the following results on the test dataset:

Test $R^2 = 0.576$

RMSE = 0.746

The R^2 score indicates that the model explains approximately 57.6% of the variation in house prices. The RMSE value of 0.746 indicates that the average prediction error is approximately \$74,600, since the target variable is measured in hundreds of thousands of dollars.

These results suggest that the linear regression model captures the main relationships between the predictors and house prices, but there is still unexplained variation because housing prices depend on many additional factors not included in the dataset.